

SOL CONSTRAINTS →

Kind of restrictions over the
inserting data into the Table.

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SOL constraints are used to specify rules for data they are used to limit the type of data that can go into a table.

This ensures that accuracy and reliability of the data in the table, if there is any violation b/w the constraint & the data action, the action is aborted.

constraints can be column level or table level constraint apply to columns.

NOT NULL = column not have null value
UNIQUE ✓

PRIMARY KEY

FOREIGN KEY

CHECK - to insert specific kind of data,

DEFAULT

CREATE INDEX - used to create index, used to retrieve data very quickly.

PRIMARY KEY CONSTRAINTS →

① It consists a unique value + not null.

② find a row uniquely.

③ It is not necessary to design in a table but it is recommended

④ There exist at most one primary key in a table

⑤ indexing is automatically done.

create table student (Roll no int(3) primary key,
name varchar(10), marks int(3), phone number(10))

Roll no	name	marks	phone
1	KUSH	98	98
2	KUCH	78	90
3	LUV	38	
4	Raj	39	

❌ Create table Student (Roll no int(3), (Enroll no int(5))
name varchar(10), marks int(3), primary
key (Roll, Enroll));

↓
table level constraint

Roll no	Enroll no	name	marks
1	10	A	40
1	20	B	50
2	20	C	60
3	30	D	70

UNIQUE KEY CONSTRAINT →

Create table Student (Roll no int(3) primary key,
name varchar(10), marks int(3), phone number(10)
unique key, (email varchar(20) unique
key);

Unique key can be many as it allow null &
distinct values also.

next row

A Create table R (A int(3), B char(2), C int(3))
unique key (A, B);

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NOT NULL CONSTRAINT →

Create table student (Roll no int(3) primary
key, Name varchar(10)
not null, marks int(3), phone number(10)
unique key not null);

CHECK CONSTRAINTS →

Create table student (Roll no int(3) primary
key, Name varchar(10)
not null, marks int(3) check (marks <= 100),
phone number(10));

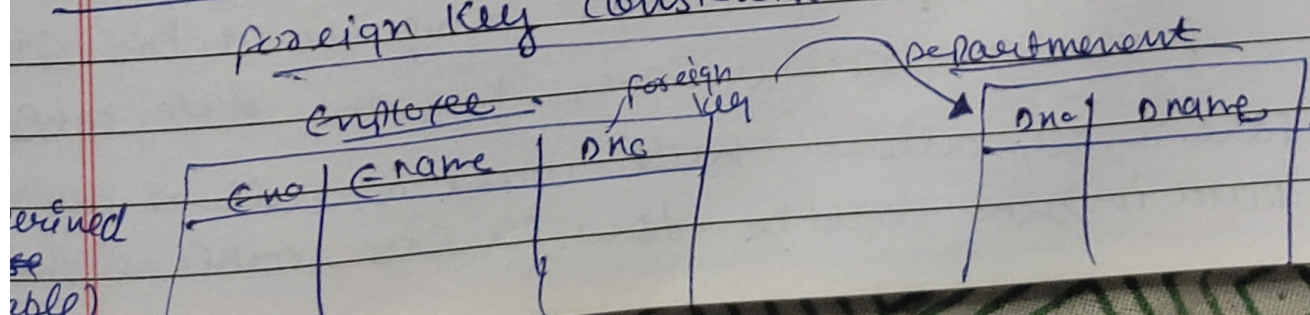
~~Create table~~ ^{Employee} (eid int(3) primary
key, salary int(5)
dept no int(2), check (salary > 10000 and
dept-no = 10));

DEFAULT CONSTRAINTS → To avoid null values.

Create table student (Roll no int(3) primary
key, Name
varchar(10), marks int(3),
phone number(10) default 9999999999);

01 CONSTRAINTS 2 →

Foreign Key Constraints -



Data should inserted in
in desired table

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Employee			Department	
EmpId	Ename	F.K	Dno.	Dname
1	KUSH	10	10	CS
2	LUV	10	20	IT
3	KIXA	20	30	EC
4			40	ME

Foreign key is repetition allowed & 42 foreign
key is 10 refer that & work allowed
not & 1

here one of emp. table is foreign key
while Dno of department is its referring

derived
emp. table is which data insert into set
base table is & not 50, 60, 70
Derived table is one field is set
into 42/1/50, 60, 70 department
table is set & 1

Foreign Key

- (1) Foreign key is implemented in derived table and
reference to Base table
- (2) more than one foreign key in table is possible
- (3) minimum one table is required to implement
foreign key
- (4) foreign key in derived table should be either
primary key or unique key of Base table
- (5) foreign key allows null value as well as duplicate
Data insertion should be done in Base table before

inserting data in derived table corresponding to foreign key

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Creation of Base table should be before derived table.

Implementation →

Create table department (d-no int(3) primary key,
dep-name varchar(20))

Create table employee (Eid int(2) primary key,
ename varchar(10),
deptid int(3), salary^{int} (5), foreign key
(dept-id) references department
(d-no));

Employee

Department

E-no	E-name	F.K d-no.	D-no	D-name
1	A	10	10	CS
2	B	10	20	EC
3	C	20	30	IT
4	D	30	40	EE

delete from derived table then no issue

but

on delete set null → derived table it null and

↓
deleting from Base table

on delete set cascade

↓
Base
table

↓
also
delete
from derived
table

on update set cascade

same update

reflected

at derived table also

on update set null

Base table

derived table null

Implementation →

Create table department (d-no int(3) primary key, d-name varchar)

Create table Employee (Emp-no int(2) primary key, Emp-name varchar(10), dep-id int(3), salary int(5), foreign key (dep-id) references department (d-no) on delete set cascade);

R

A	B
1	2
2	3
3	4
4	5
5	1

Foreign key (on delete cascade)
let (1,2) deleted.
base table
derived table

all references table get null

let on delete not
we apply delete on
as (1,2) so,

there is delete of
element (only
not all row table)

while in
on delete cascade
all reference
related to (1,2) row
get deleted
rows in this
table get null

ea

R F.K

A	B
1	2
2	3
3	1
4	4
5	5
2	5

on table
Railroad

S F.K

C	D
2	1
3	2
1	2
4	3
5	3
5	5

on table
Passenger

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ed (2,1)
be deleted

SOL CONSTRAINTS PART-3 →

Create index id_name on Employee (ename);

→ It allows duplicate value

Create unique index id_name on Employee (ename, last);

↓ not allow duplicate value

DEFINE CONSTRAINTS USING ALTER TABLE COMMAND →

Alter table student Add primary key (roll);

Alter table student modify roll int(5) primary key;

we can Add any type of constraint using alter command

Alter table Employee Add foreign key (dept-no) references department (dept-no);

DROPPING CONSTRAINTS USING ALTER TABLE COMMAND

Alter table student drop primary key;

Alter table student add check (marks <= 100)

1 Agas date C=100
wala etc